

AI-Powered Resume Analysis and Career Recommendation Systems: A Review

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Abstract:

In modern recruitment, automated systems analyze resumes and match candidates to job descriptions using Artificial Intelligence (AI) and Natural Language Processing (NLP). Traditional Applicant Tracking Systems (ATS) rely heavily on keyword matching and often overlook semantic understanding, reducing effectiveness. Recent studies propose advanced methods using deep learning and transformer embeddings to improve resume parsing, matching, and recommendation accuracy. This review paper analyzes recent research on automated resume screening, semantic matching, and job recommendation frameworks, identifies key strengths and limitations, and highlights future research directions.

Keywords: Resume Parsing, Applicant Tracking System, Natural Language Processing, Machine Learning, Semantic Matching, Career Recommendation

1. Introduction

Efficient resume screening and job matching are critical challenges in talent acquisition. Manual screening is time-consuming and prone to bias. Traditional ATS algorithms filter resumes using simple keyword checks and structured fields but lack deep semantic understanding. AI and NLP techniques enable automated extraction of candidate features and semantic comparison with job requirements, improving scalability and fairness. This paper reviews state-of-the-art research on automated resume analysis frameworks, their techniques, and performance trends.

2. Background and Motivation

Applicant Tracking Systems automate resume screening by parsing applicant documents and matching them with job descriptions. Traditional ATS rely on keyword search, but struggle with unstructured resumes and contextual meaning. Advances in NLP and machine learning enable structured information extraction and semantic representation of candidate profiles.

3. Literature Review

A. Automated Resume Parsing and Feature Extraction

- **NLP-Based Resume Parsing:** Lokhande et al. propose an AI-based resume analyzer that uses NLP to extract structured information and generates semantic similarity scores between resumes and job



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AI-Powered Resume Analyzer & Career Recommendation System

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Abstract:

In today's competitive job market, a significant number of resumes are filtered out by Applicant Tracking Systems (ATS) before reaching recruiters. Many students and fresh graduates lack awareness of required skills, resulting in poor job matching and low employability. This paper presents an AI-Powered Resume Analyzer and Career Recommendation System that leverages Natural Language Processing (NLP) and Machine Learning (ML) techniques to evaluate resumes, calculate ATS compatibility, identify skill gaps, and provide personalized career guidance. The system integrates job, course, and project recommendation APIs along with an AI-based chatbot for real-time assistance. The proposed solution improves resume quality, enhances interview readiness, and supports data-driven career decisions.

Keywords: Resume Parsing, Applicant Tracking System, Natural Language Processing, Machine Learning, Semantic Matching, Career Recommendation

1. Introduction

The recruitment process has evolved significantly with the adoption of automated systems such as Applicant Tracking Systems (ATS). These systems filter resumes based on keyword matching and formatting, leading to the rejection of many candidates before human evaluation. Students often lack clarity regarding resume optimization, required skills, and job readiness.

This research proposes an intelligent system that analyzes resumes using AI techniques and provides actionable insights including ATS scores, job match percentages, and skill gap analysis. Additionally, it offers recommendations for jobs, courses, and projects, along with chatbot-based career guidance.

2. Literature Review

Previous studies have explored resume parsing and job recommendation systems using NLP techniques. Jurafsky and Martin (2023) highlighted the importance of NLP in text analysis. Various research works have implemented resume classification and keyword extraction using machine learning algorithms.

However, most existing systems focus only on resume screening and lack comprehensive career guidance features. The proposed system bridges this gap by integrating resume analysis with personalized recommendations and AI chatbot support.

3. Problem Statement

- A large number of resumes are rejected due to poor ATS compatibility.